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Dimensionality Reduction

A first taste of unsupervised

A lesser evil

- ❖ A perfect model is complete
- ❖ We are in the useful model business



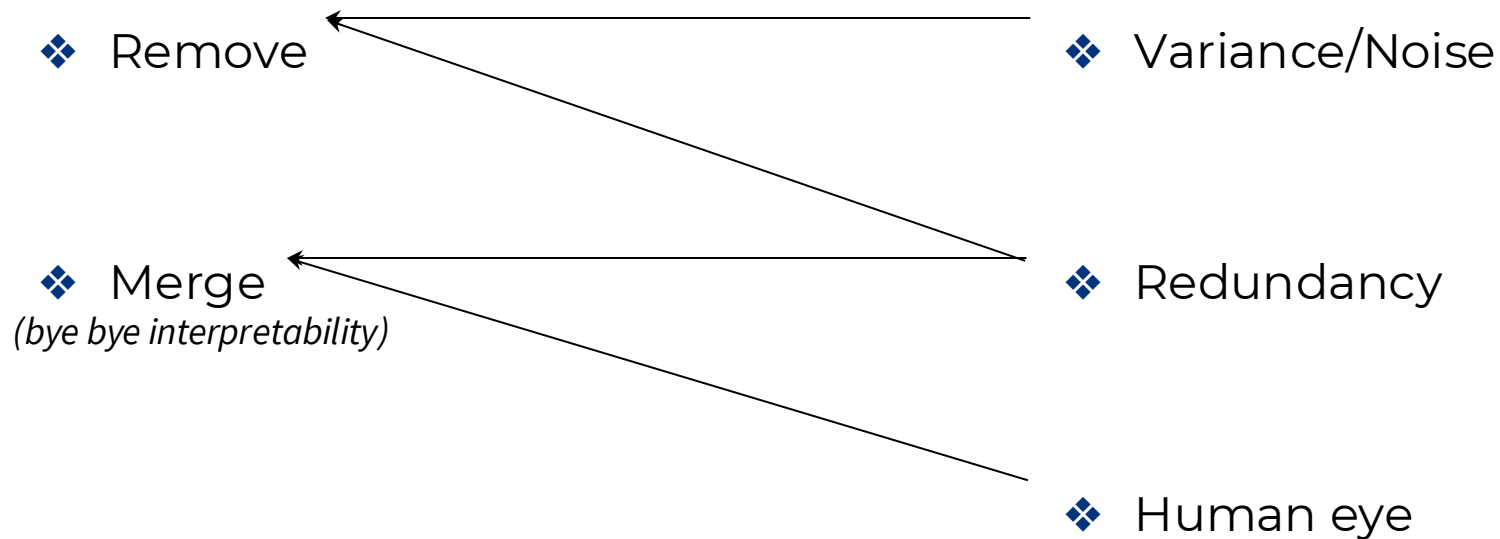
- ❖ Which dim is best?

High dimensional spaces are evil

- ❖ Everything can be explained (spurious)
- ❖ Sparsity (exponential)
- ❖ Everything is similar & different to everything (distances)
- ❖ No free lunch (opinions)
- ❖ If you can't beat them, do something different (enrich)

2 paths

2 motivations



Neighbors vs Neighbourhoods

- ❖ 12,000 (earth diameter, ED)
- ❖ 384,000 (earth-moon EM) (x32 ED)
- ❖ 150,000,000 (earth-sun) (x390 EM) (x12,500 ED)



- ❖ **Global:** Relative location wrt far away elements (global, major trends)
- ❖ **Local:** Relative location wrt near by elements (fine-grained, intricate)

LDA



Fisherfaces

❖ Linear Discriminant Analysis

- Supervised: Requires categorization of data (labels! global!)
- Creates new dimensions based on linear combinations of old ones
- Optimize with Fisher's criterion
 - Minimize variance of within-class, maximize between-class means dist.
- Assum. Gaussianity. Sensitive to outliers. Sensitive to imbalance.
- Likely interpretable

PCA

❖ Principal Component Analysis

- Keep max. data variance (no labels)
- Linear and kernel PCA
- Produces uncorrelated features
- Rank of features. Interpretability?
- Sensitive to outliers
- Keeps global structure
- Does not assume Gaussianity [13,17]



Eigenfaces

MDS

❖ Multi Dimensional Scaling

- Optimize stress criterion: Keep pairwise distances (global)
- Distance matrix. No labels.
- Non-linear transforms
- Interpretability based on similarity metric.
- Does not assume Gaussianity
- Sensitive to outliers

t-SNE

- ❖ t-Distributed Stochastic Neighbor Embedding
 - Non-linear transforms
 - Local (perplexity, num. nearest neighbours)
 - Outlier sensitivity
 - Distance matrix. No labels.
 - Does not assume Gaussianity

U-Map

❖ Uniform Manifold Approximation and Projection

- Divergence between pairwise similarities in the high-dim. vs low-dim. space
- Distance matrix. No labels.
- Non-linear transforms
- Does not assume Gaussianity
- Local. Num. Neighbors and Min Distance to globalize
- Robust to outliers

[13,17]

Others

❖ TriMAP

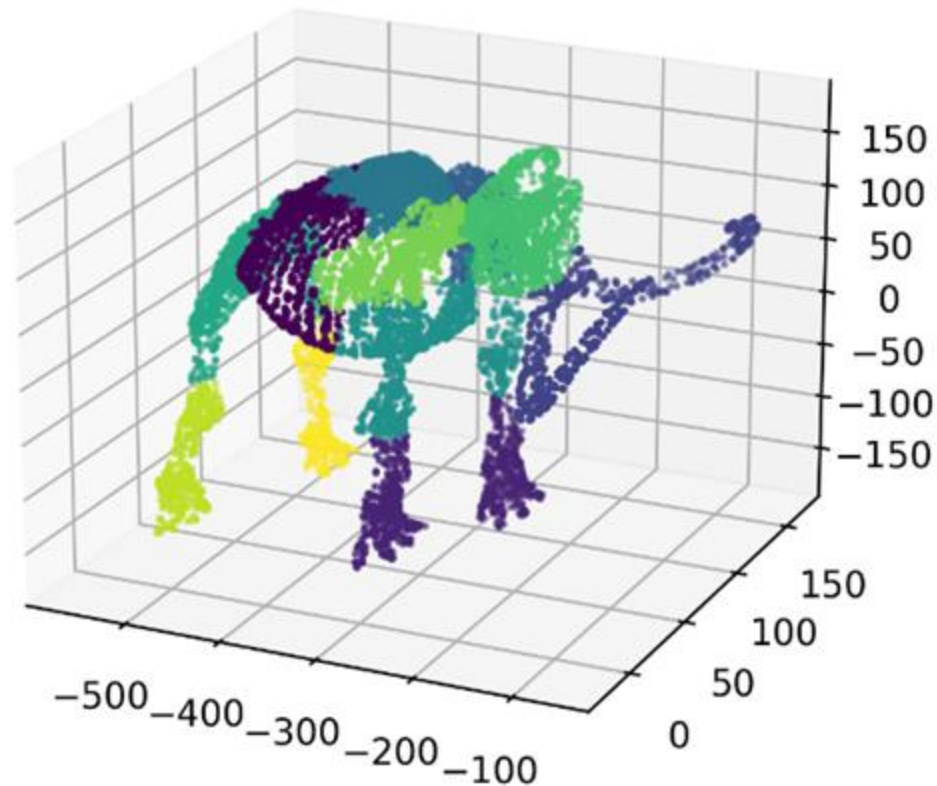
- Target triplet relations
- One close, one far

❖ PaCMAC

- Varying the neighborhood sizes for different regions.

Mammoth

- ❖ Local?
- ❖ Global!

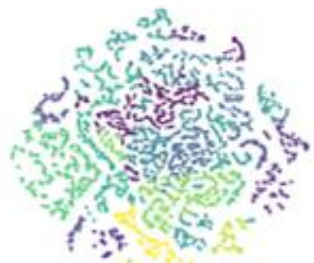


Mammoth: MDS

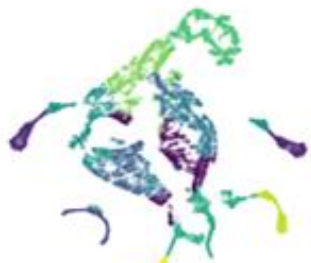
❖ Global 2D



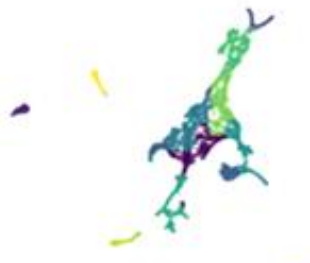
t-SNE(perplexity=10)



t-SNE(perplexity=125)



LargeVis(perplexity=125)



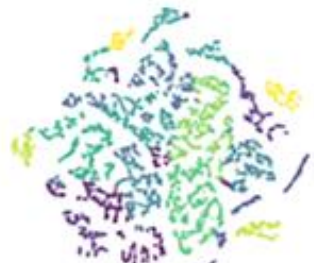
UMAP(n_neighbors=10)



TriMap(n_inliers=10)



t-SNE(perplexity=20)



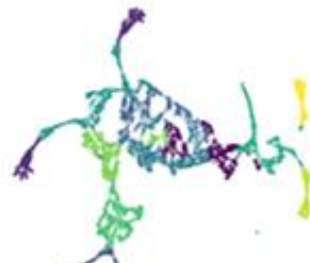
t-SNE(perplexity=250)



LargeVis(perplexity=250)



UMAP(n_neighbors=20)



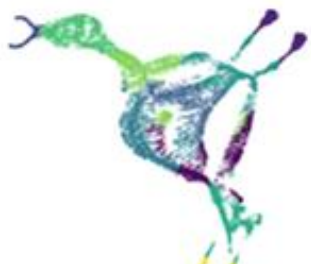
TriMap(n_inliers=20)



t-SNE(perplexity=40)



t-SNE(perplexity=500)



LargeVis(perplexity=500)



UMAP(n_neighbors=40)

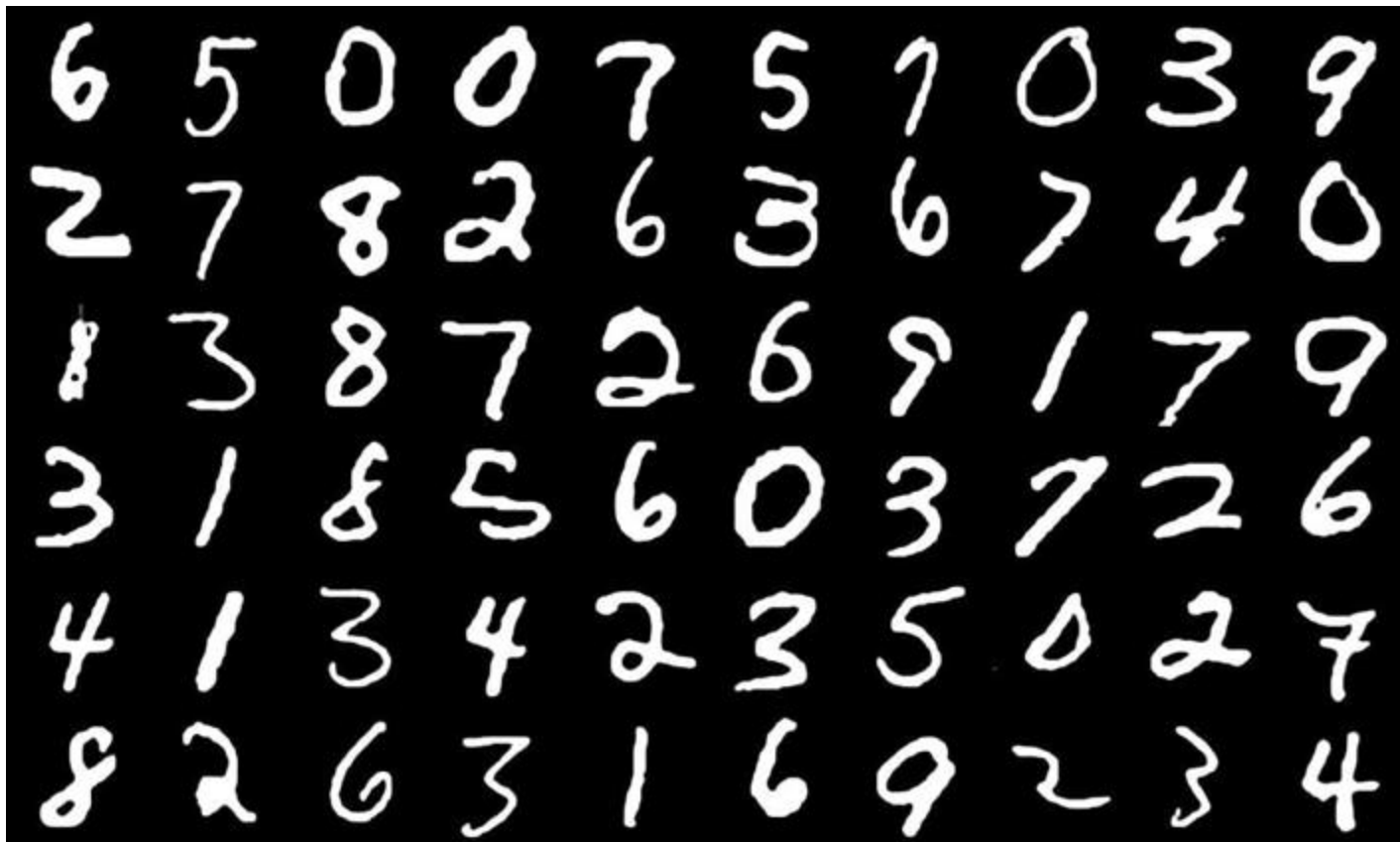


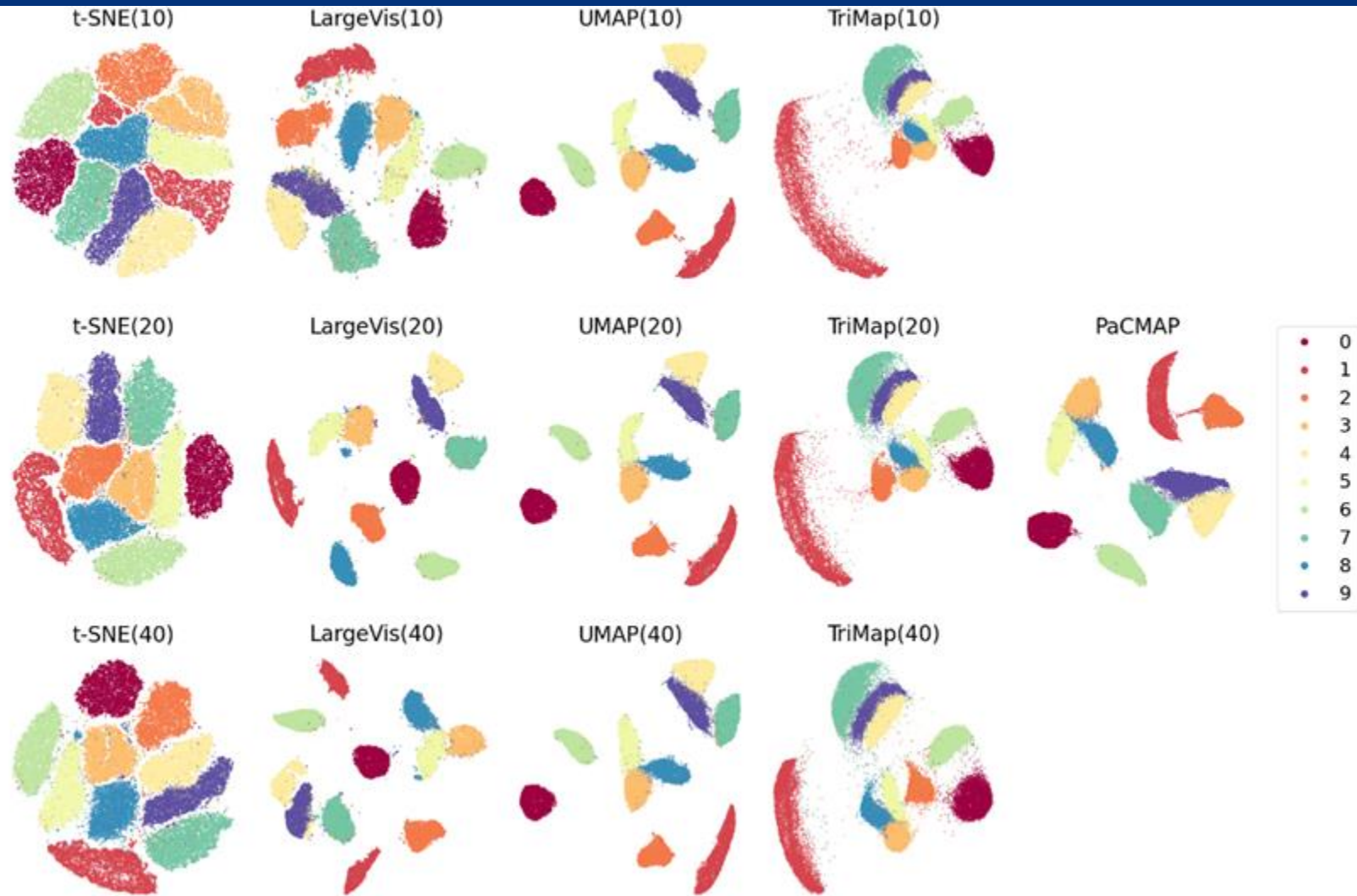
TriMap(n_inliers=40)



One non-random example #2

- ❖ MNIST
- ❖ Global?
- ❖ Local!





MDS: Global

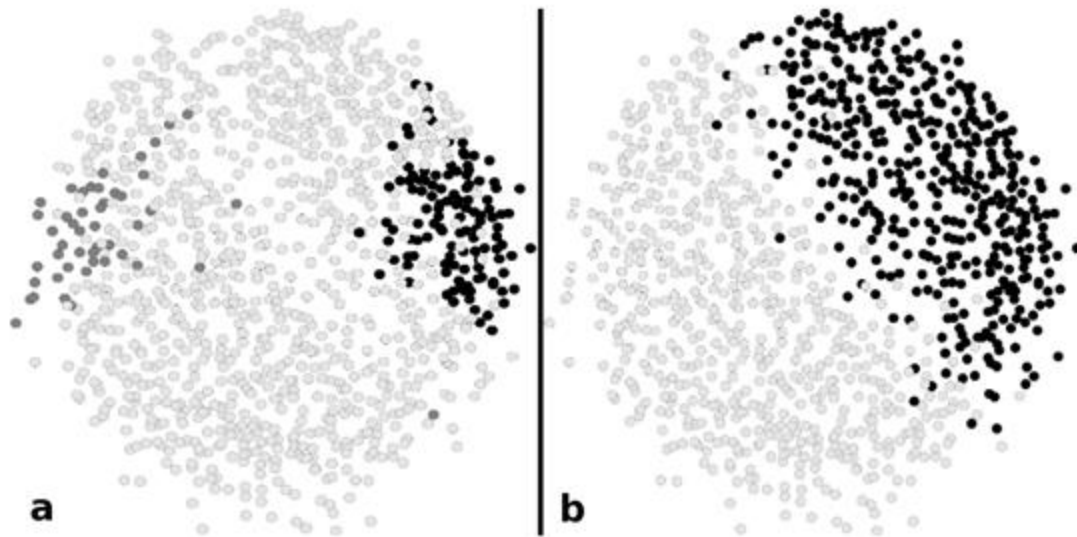


Figure 4: Scatter plot of image class vector similarities built through metric multi-dimensional scaling. (a) black circles belong to images labelled as hyponyms of synset "dog, domestic dog, *Canis familiaris*" and dark grey circles belong to images labelled as hyponyms of synset "wheeled vehicle". (b) black circles belong to images labelled as hyponyms of synset "living thing, animate thing".

- ❖ Non-liv False Pos.
 - Lobster
 - Baseball player
 - Dragonfly
- ❖ Liv False Pos.
 - Snorkel
 - Dog-Sled
 - Coral reef

The hard part

- ❖ There is no right one
- ❖ Human bias
- ❖ Tiny random things meets pareidolia
- ❖ Always, but never only



An exercise

- ❖ Task: Predict how many accidents will a car have this year
- ❖ List 10 variables of interest. For each:
 - Type, distrib., outliers, pre-processing
- ❖ Data from city A, insurance company X & Y. City B, company X.
 - Propose two data splits for two purposes